



DairyDesk

Complete Setup Guide — From Zero to Running the App

macOS (Apple Silicon & Intel)

PostgreSQL 16

Django + DRF

Vite + React

Docker

For a fresh Mac with **none** of the required software installed.

Repository: github.com/TanmayKamble004/DairyDesk · Prepared July 2026

Who this is for. You have a Mac with nothing installed yet. Follow the sections **top to bottom** — don't skip ahead. Part A installs the tools the project needs; Part B gets the code and runs it. Total time: roughly **30–45 minutes**, most of it downloads.

What you'll end up with

Piece	What it is	Runs at
Database	PostgreSQL 16 inside Docker	<code>localhost:5432</code>
Backend	Django REST API (the server)	<code>localhost:8000</code>
Frontend	React app (what you look at)	<code>localhost:5173</code>

How to open a terminal: Press `⌘ + Space`, type **Terminal**, and hit `Return`. You'll use this a lot. Your Mac uses the **zsh** shell by default.

Part A — Install the required software

We'll install **Homebrew** first — the standard macOS package manager — then use it to install Git, Python, and Node in one line each. Docker Desktop is a normal app download. After each install, run the verify command.

0 Homebrew — the package manager

Paste this into Terminal and press `Return`. It will ask for your Mac password (typing is invisible — that's normal) and may take a few minutes:

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
```

Apple Silicon (M1/M2/M3/M4): when Homebrew finishes it prints two `Next steps` lines starting with `echo` and `eval`. **Copy and run those two lines** so the `brew` command works, then **close and reopen Terminal**. (On Intel Macs this isn't needed.)

Verify:

```
brew --version
```

1 Git — downloads the code

```
brew install git
```

Verify:

```
git --version
```

macOS may already have a basic Git. If `git --version` already works, you can skip this step — but `brew install git` gives you an up-to-date version.

2 Python 3.12+ — runs the backend

```
brew install python@3.12
```

Verify:

```
python3 --version
```

You should see `Python 3.12.x` (or newer).

On macOS the command is `python3` (and `pip3`), not `python`. Use `python3` everywhere in this guide.

3 Node.js 20 LTS — runs the frontend

```
brew install node@20
```

Verify:

```
node --version  
npm --version
```

`node` should be `v20.x` or newer, and `npm` should print a version too.

If `node` isn't found after installing, Homebrew printed a line to add it to your PATH. Run the `echo ... >> ~/.zshrc` line it showed (or use the non-versioned `brew install node`), then reopen Terminal.

4 Docker Desktop — runs the database

Download the correct build for your Mac from <https://www.docker.com/products/docker-desktop/>:

- ☐ **Apple Silicon** (M1/M2/M3/M4) → “Download for Mac – Apple Chip”
- ☐ **Intel** → “Download for Mac – Intel Chip”

Not sure which? Click the menu (top-left) → **About This Mac**: “Apple M...” = Apple Silicon.

Open the downloaded `.dmg` and **drag the Docker whale into Applications**. Then:

- ☐ Launch **Docker** from Applications (or Launchpad). Approve the permission prompt / enter your password if asked.
- ☐ Wait until the whale icon in the **menu bar** (top-right) stops animating and shows “**Engine running**”. First launch can take a minute or two.
- ☐ You can skip / close the sign-in — an account is not required.

Verify (with Docker running):

```
docker --version
docker compose version
```

Checkpoint. All of these should print a version with no errors: `brew` , `git` , `python3` , `node` + `npm` , and `docker` . If any fail, close every Terminal window, reopen a fresh one, and try again — new installs only show up in newly opened terminals.

Part B — Get the code and run it

1 Get access & clone the repository

DairyDesk is a **private** repository, so Tanmay must add you as a collaborator on GitHub first (you'll get an email invite — accept it). Then, in Terminal, pick a folder for the project and clone it:

```
# Go to your Documents folder (or wherever you want the project)
cd ~/Documents

# Download the code
git clone https://github.com/TanmayKamble004/DairyDesk.git
cd DairyDesk
```

Sign-in prompt: the first time you clone, Git asks you to authenticate to GitHub. Sign in with the account that received the invite. If it fails, install the GitHub CLI (`brew install gh`) and run `gh auth login` once, then retry the clone.

2 Create the environment files

The app reads its settings from `.env` files. Copy the provided templates — the default values work for local development as-is:

```
cp .env.example .env
cp frontend/.env.example frontend/.env
```

3 Start the database (Docker)

Make sure **Docker is running** (steady whale in the menu bar). Then, from the `DairyDesk` folder:

```
docker compose up -d
```

This downloads and starts PostgreSQL 16 in the background. Confirm it's healthy:

```
docker compose ps
```

Wait until the `db` row shows **(healthy)** — a few seconds after first start.

4 Set up and start the backend (Django)

Still in the `DairyDesk` folder, move into the backend, create an isolated Python environment (a “virtual environment”), and activate it:

```
cd backend
python3 -m venv .venv
source .venv/bin/activate
```

When activated, your prompt shows `(.venv)` at the start of the line.

Now install the backend's Python packages, set up the database tables, load demo data, and start the server:

```
pip install -r requirements.txt
python manage.py migrate
python manage.py seed_demo
python manage.py runserver
```

Inside an activated venv, `python` and `pip` (no "3") correctly point to your venv — so the commands above are right. Outside the venv, keep using `python3` / `pip3`.

When you see **"Starting development server at http://127.0.0.1:8000/"**, the backend is up.

Leave this terminal running. Closing it stops the backend. You'll open a **second** terminal for the frontend in the next step.

5 Set up and start the frontend (React)

Open a **new** Terminal window ( + N), leaving the backend one running. Go to the project's frontend folder and start it:

```
cd ~/Documents/DairyDesk/frontend
npm install
npm run dev
```

`npm install` downloads the frontend packages (takes a minute the first time). When you see a **Local: http://localhost:5173/** line, it's ready.

6 Open the app & log in

In your browser, go to <http://localhost:5173>. Log in with one of the demo accounts created by `seed_demo`:

Role	Username	Password	Access
Owner	<code>owner</code>	<code>owner123</code>	Full access + Django admin
Staff	<code>staff</code>	<code>staff123</code>	Limited access

 **That's it — DairyDesk is running.** Explore the 3D inventory shelf, products, customers, orders and invoices. `seed_demo` is safe to re-run anytime to reset the demo data.

Running it again next time

Once everything is installed, you **don't** repeat Part A. To start the app on any later day:

#	Do this	Where
1	Open Docker and wait for "Engine running"	Applications / Launchpad
2	<code>docker compose up -d</code>	Terminal in DairyDesk
3	<code>cd backend</code> → <code>source .venv/bin/activate</code> → <code>python manage.py runserver</code>	Terminal 1
4	<code>cd frontend</code> → <code>npm run dev</code>	Terminal 2 (new)
5	Open http://localhost:5173	Browser

To stop everything: press **Control + C** in both terminals, then run `docker compose down` to stop the database. Your data is preserved for next time.

Troubleshooting

Symptom	Fix
"Cannot connect to the Docker daemon" / <code>docker</code> errors	Open the Docker app and wait for the menu-bar whale to go steady ("Engine running"), then retry.
Backend: database connection refused	The DB wasn't ready. Run <code>docker compose ps</code> and wait for (healthy) , then start the backend again.
" <code>command not found: brew</code> " after installing Homebrew	On Apple Silicon you must run the two Next steps lines Homebrew printed (the <code>echo</code> + <code>eval</code> lines), then reopen Terminal.
" <code>command not found: python</code> "	On macOS use <code>python3</code> (and <code>pip3</code>) — except inside an activated venv, where plain <code>python</code> works.
" <code>node</code> / <code>npm</code> not found" after brew install	Run the PATH line Homebrew printed (<code>echo ... >> ~/.zshrc</code>), then reopen Terminal — or reinstall with plain <code>brew install node</code> .
"Port already in use" (8000, 5173, or 5432)	Another program is using that port. Close it, or restart your Mac and try again.
<code>npm install</code> fails partway	Delete the <code>frontend/node_modules</code> folder and run <code>npm install</code> again on a stable connection.
Login fails / no demo accounts	In the backend terminal (venv active), re-run <code>python manage.py seed_demo</code> .